Real-World Coordinate-Plane Problems; Perimeter

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lesson 6 — Real-World Coordinate-Plane Problems; Perimeter	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.2

2. Standards

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Must. Find the perimeter of a figure on the coordinate plane, each slanted side by its right triangle (a triangle with no right angle split by its height); the straight-line and along-the-grid distances on a map, and how much shorter the straight line is; an irrational total kept exact and rounded once.

Must not. The distance formula (Clarification 2); radicands over 225 or with a square factor; combining radicals (√34 + √34 is two lengths added, not 2√34 — Algebra 1).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I and III: one perimeter as sides on the grid, as right triangles and as a sum
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Notes II and boards 5, 7 and 8: maps, the streets against the straight line

4. Learning Target and Essential Question

Learning target. I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

Essential question. How do you measure around a shape whose sides do not follow the grid lines?

Building on. 6.05 — the distance between two points on the grid; 6.03 — the Pythagorean Theorem; Grade 6 (MA.6.GR.1.3) — the perimeter of a rectangle on the coordinate plane.

Working toward. 6.07–08 — similar triangles; Geometry — perimeters on the plane, and the distance formula, which is this triangle written as one line.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is the straight path shorter than the streets unless the two points share an x or a y?	DOK 2
Notes III rounded once, at the end. When does rounding each side first change the answer?	DOK 3
A triangle has no right angle. Where would you draw a line to give it two?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$. Find its perimeter.	***30 units***	17 — left the slanted side out [Notes I: every side]; 34 — took AC as 12 + 5 [MA.8.GR.1.2: the theorem, not a sum]
2	A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$. Find its perimeter, to the nearest tenth.	***22.8 units***	13 — left the slanted side out [Notes I]; 22.9 — rounded 9.849 to 9.85 and then to 9.9 [one rounding, at the end — Notes III]; 26 — the slanted side as 4 + 9 [MA.8.GR.1.2]
3	Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle. Find its perimeter.	***42 units***	40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]; a perimeter with 12 in it — the height is inside the triangle, not a side [B1G-M, GR.1.2: the theorem on a triangle with no right angle]
4	What is the perimeter of the triangle with corners (−2, 3), (2, 3) and (2, −4), to the nearest tenth?	***D — 19.1 units***	(A) 4 + 7: left the slanted side out [Notes I: every side]; (B) 4 + 7 + 11: the slanted side taken as the sum of the legs [MA.8.GR.1.2: the theorem, not a sum]; (C) 4 + 7 + 65: forgot the square root [B1G-M, GR.1.1: root errors]
5	On a map in blocks, the library is at $(-7, -1)$ and the pool at $(5, 4)$. How much shorter is the straight line than the streets?	***4 blocks shorter***	13 or 17 — one path, not how much shorter [the question asks a difference]; 30 — added the two paths [the question asks a difference]; 2 across, from 7 − 5 [Florida B1G-M, GR.1.2: integer errors across an axis]
6	How long is a diagonal of the rectangle with corners (−3, −4), (6, −4), (6, 8) and (−3, 8)?	***A — 15 units***	(B) 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]; (C) 9 + 12 + 9 + 12: the perimeter, not one diagonal [Notes II: the straight line is one segment]; (D) 6 and 8, one corner's coordinates, taken as the legs [Math Nation names wrong horizontal and vertical lengths]
7	On a park map in meters, the fountain is at $(-5, 2)$ and the gate at $(4, -3)$. How far is it straight across, to the nearest tenth?	***$\sqrt{106} \approx 10.3$ m***	14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]; √26 — 4 − 5 = 1 for the across leg [Florida B1G-M, GR.1.2: integer errors across an axis]
8	On a trail map in miles, the camp is at $(-3, -6)$, the lake at $(2, 6)$ and the trailhead at $(-1, 0)$. How far is the camp from the lake, in a straight line?	***13 mi***	17 — added the legs [MA.8.GR.1.2]; a distance to the trailhead — the wrong pair of points

#	What the board asks	Answer	If the board is wrong
9	Mia says the segment from (0, 0) to (3, 1) is 3 units long, because it crosses 3 grid squares. Is she right? Explain, and find its length to the nearest tenth.	***No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.***	'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]; 4 — added the legs [MA.8.GR.1.2]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. A printed grid for every board. Find the sides on grid lines first and write them on the figure; then the slanted side, by its triangle. "Every side counts" said aloud with every perimeter.

ELL. **Perimeter**: around the outside — every side. **Along the streets**: on the grid lines. **Straight line**: one segment, as a bird flies. Trace each with a finger on the grid.

Enrichment. Plot (0, 0), (6, 0) and (3, 4). Is it a right triangle? (No: 25 + 25 is not 36.) Find its perimeter. (5 + 5 + 6 = 16.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Mia counted the squares her segment crosses, but a slanted segment is the hypotenuse of a right triangle — legs 3 and 1, √10 ≈ 3.2, longer than 3. A board that says only '3.2' has the length and not the reason.

Vocabulary. **perimeter**; **height of a triangle**; **straight-line distance**

Materials. Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Pythagorean theorem: find the perimeter — VGE.